Course Name: Neural Networks
Course code: AI303
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Fall 2025 – 2026

AI BloodScan

Members Names and Role

	Name	ID	Role
1	Ashraf amr	202300962	Mobile app and integration with backend
2	Kareem abd el nasser	202303078	API,GUI, and mlp model
3	Yousef ali elfar	202302115	Mlp and backend
4	Ezz eldin yasser	202303039	Preprocessing and collecting dataset
5	Sandra Samer Samir	202303334	CNN + Tabnet
6	Rosana Zarif	202303306	CNN + Tabnet
7	avronia magdy	202303908	comparison and documentation +Tabnet

1) Introduction:

Project Idea & Objective

The project aims to develop an automated system capable of extracting, analyzing, and classifying Complete Blood Count (CBC) test results directly from laboratory report images. The system uses OCR to read values and a machine learning model to classify each blood metric as Normal or Abnormal based on medically verified reference ranges.

The main objective is to help society by enabling faster, more accessible, and more accurate preliminary health assessments—especially for people who cannot interpret medical results or in emergency scenarios where quick triage is essential.

Dataset Type & Specifications

1. Type of Dataset:

We used two types of datasets in our project:

- **The first type was tabulated data in a CSV file, which was numerical data. Only the label feature was strange, and we processed it (This was for training the model)**
- **The second type was actual images from reports from different labs, (and this was for final testing of the entire application)**

The dataset used in this project is a tabular numerical dataset.
Originally, the data comes from images of CBC (Complete Blood Count) laboratory reports, which are processed using OCR.
After OCR extraction and cleaning, the final dataset becomes:

- **Structured numerical data (tabular format)**
- **Each row represents one patient's CBC test results**
- **Each column represents a specific blood measurement (e.g., Hb, WBC, RBC)**

So the raw data is images, but the processed dataset used for model training is tabular numeric data.

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2. Dataset Description

- **Overall Size:** `(7196, 17)`

After cleaning, preprocessing, and balancing:

- **Total samples:**

```
Shapes after split:

Train.csv: (5037, 16)

Val.csv: (1079, 16)

Test.csv: (1080, 16)
```

- **Each sample contains: 17 numerical features (Hb, RBC, WBC, PLT, HCT, MCV, MCH, MCHC, RDW, MPV, PCT, etc.)**

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3. Number of Classes

The classification task contains 15 classes:

1. **0 → Anemia of Chronic Disease**
2. **1 → Bacterial Infections**
3. **2 → Beta-Thalassemia Minor**
4. **3 → Chronic Leukemias**
5. **4 → Infectious Mononucleosis**
6. **5 → Iron Deficiency**
7. **6 → Leukocytosis**
8. **7 → Leukopenia**
9. **8 → Microcytic Anemia**
10. **9 → Normal**
11. **10 → Normocytic Anemia**
12. **11 → Polycythemia Vera**
13. **12 → Thrombocytopenia**

14.13 → Thrombocytosis
15.14 → Viral Infections

Each sample is labeled based on medical reference ranges after normalization.

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4. Size per Class

Because medical datasets are usually imbalanced, the original distribution looked like:

```
df['Diagnosis'].value_counts()
```
✓ 0.4s

Diagnosis	# count
Anemia of Chronic Disease	1355
Normocytic Anemia	1209
Bacterial Infections	1058
Thrombocytosis	619
Beta-Thalassemia Minor	429
Normal	415
Infectious Mononucleosis	380
Viral Infections	323
Leukopenia	233
Iron Deficiency	228

To handle imbalance, techniques like SMOTE and Weighted Sampling were applied.

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5. Feature Specifications

Each sample includes:

- **Pure numerical values**
- **All features normalized and cleaned from OCR noise**
- **Examples of feature list:**
  - **Hemoglobin (Hb)**
  - **Red Blood Cells (RBC)**
  - **White Blood Cells (WBC)**
  - **Platelets (PLT)**
  - **Hematocrit (HCT)**
  - **MCV, MCH, MCHC, RDW, MPV, PCT**
  - **(depends on available values per report)**

Total features: input_dim (14)

6. Source of Data

- **Extracted from CBC report images via a custom OCR pipeline.**
- **OCR → data cleaning → unit normalization → final tabular dataset.**

2) Challenges in the dataset (including pre-processing phase):

1. Lack of Standardization in CBC Reports

- **CBC reports differ significantly between laboratories.**
- **Variation in table layouts, fonts, naming conventions (e.g., HGB vs Hb), and units.**
- **This inconsistency made it difficult to create a unified parsing strategy.**

2. OCR Instability Due to Image Quality

- **Many images had poor lighting, shadows, blur, or skew.**
- **OCR models struggled with handwritten notes or low-contrast prints.**
- **Numbers were often misread: (1→7), (0→O), (3→8), or missing decimal points.**

3. Missing and Incomplete Values

- **Some CBC sheets had missing rows or cropped images.**
- **OCR occasionally skipped values or misaligned them with wrong test names.**
- **Required building rules to detect and handle missing values safely.**

4. Unit Variability Across Labs

- **Example: WBC may appear as 6.2, 6200, or 6×10^3.**
- **Required a normalization logic to scale all units to a single standard.**

- **Without unit-unification the dataset became unreliable.**

5. Presence of Outliers

- **Due mostly to OCR errors rather than real medical values.**
- **Unrealistic values like:**
  - **WBC = 60000**
  - **Hb = 140**
  - **Platelets = 8**
- **Required implementing sanity checks and outlier filtering rules.**

6. Inconsistent Naming of Features

- **Example variations:**
  - **Hemoglobin = Hb / HGB / Haemoglobin**
  - **Platelets = PLT / Platelet Count**
- **Required feature-mapping to unify the dataset.**

7. Noisy OCR Text

- **OCR sometimes output words instead of numbers.**
- **Preprocessing had to clean: extra symbols, commas, units, trailing text.**
- **Required regex-based cleaning + validation logic.**

8. Imbalanced Data Distribution

- **Majority of records were Normal, very few Abnormal samples.**
- **Caused ML models to overfit and predict "Normal" for every input.**
- **Required applying resampling or class-weight adjustments.**

```
df['Diagnosis'].value_counts()
✓ 0.4s
```

Diagnosis	# count
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Normal	415
Infectious Mononucleosis	380
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2) Pre-processing Methodologies Used

1. OCR Post-processing Pipeline

To fix misread values and normalize extracted text.

Techniques used:

- **Regex-based number extraction**
- **Decimal reconstruction (e.g., converting 34 back to 3.4 when expected)**
- **Unit normalization (×10³ → conversion)**
- **String similarity for feature name matching**

Pseudo Code Example:

```
function clean_ocr_value(text):
    value = extract_numeric(text)
    if missing_decimal(value) and expected_range_small:
        value = insert_decimal(value)
    if "x10^3" in text or "×10³" in text:
        value = value * 1000
    return value
```

Why used:

- **OCR output is unreliable and requires strict rule-based correction.**

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2. Feature Normalization & Mapping

Unifying all column names.

Pseudo Code:

```
mapping = {
  "HGB": "Hemoglobin",
  "Hb": "Hemoglobin",
  "Haemoglobin": "Hemoglobin"
}

function normalize_feature(name):
    return mapping.get(name, name)
```

Why used:

- **Consistency is necessary for ML model training.**

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3. Outlier Detection (Rule-based)

Used because medical ranges are well-defined.

Rules:

```
if value < min_allowed or value > max_allowed:
    mark_as_outlier()
    replace_with_null()
```

Why used:

- **Outliers were almost always from OCR mistakes.**

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4. Handling Missing Data

Methods used:

- **Mean/median imputation**
- **Predictive imputation for correlated values**
- **Complete-case removal when safe**

Pseudo Code:

```
for each feature in dataset:
    if value is null:
        value = median(feature)
```

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5. Class Imbalance Techniques

To fix Normal >> Abnormal imbalance.

Methods used:

- **Oversampling (SMOTE)**
- **Undersampling**
- **Class-weighted loss in the ML model**

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6. Neural Network Model (MLP)

Used for classification.

Why MLP?

- **Tabular data**
- **Lightweight**
- **Good baseline**

Simplified Code Module:

```
class MLP(nn.Module):
    def __init__(self, input_dim, output_dim):
        super().__init__()
        self.layers = nn.Sequential(
            nn.Linear(input_dim, 256),
            nn.ReLU(),
            nn.Dropout(0.3),
            nn.Linear(256, 128),
            nn.ReLU(),
            nn.Linear(128, output_dim)
        )

    def forward(self, x):
        return self.layers(x)
```

Why this architecture:

- **Depth is enough for feature interactions.**
- **Dropout reduces overfitting from small dataset.**

3) Processing Phase:

We used an MLP (Multi-Layer Perceptron) architecture because:

- **Input data is numerical (not image pixels or text sequences).**
- **Each CBC metric is treated as an independent numeric feature.**
- **CNN and LSTM are unnecessary because we are not working with spatial or sequential patterns.**

Architecture (Simplified):

The MLP consists of:

1. **Input Layer**
    - **Takes all CBC numerical features (e.g., Hb, WBC, Platelets, etc.) as inputs.**
2. **Hidden Layer 1**
    - **256 neurons**
    - **Activation: ReLU**
    - **Batch Normalization to stabilize training**
    - **Dropout to reduce overfitting**

3. **Hidden Layer 2**
    - **128 neurons**
    - **ReLU + BatchNorm + Dropout**
4. **Hidden Layer 3**
    - **64 neurons**
    - **ReLU + BatchNorm + Dropout**
5. **Output Layer**
    - **Number of neurons = number of classes (Normal / Abnormal)**
    - **Outputs raw logits for classification**

Why This Architecture?

- **Works very well with structured medical data (numerical features)**
- **Dropout + BatchNorm help control overfitting because dataset is relatively small**
- **Deep enough to learn non-linear relationships between CBC measurements**
- **Lightweight and efficient for deployment on CPU or mobile devices**

This provides a lightweight, fast model suitable for deployment in mobile systems.

4) Comparisons:

1. Comparison with Other Architectures

- **MLP vs CNN:**
  CNN excels in images; our data is already numerical → CNN unnecessary.

- **MLP vs LSTM:**
  LSTM models temporal sequences; CBC does not rely on time patterns.

Category	**Aspect**	**Model 1: Hybrid 1D-CNN + Residual + Self-Attention**	**Model 2: TabNet (Feature Tokenizer + Transformer)**	**Winner & Reason (on your blood-parameter dataset)**
Architecture Type	Core design	Treats 16 tabular features as a fixed-length 1D sequence → Conv1D layers → Residual blocks → Global Multi-Head Self-Attention → Classification	Feature tokenizer (embeddings) → Sequential attentive decision steps with Sparsemax masks → No positional assumption	TabNet – Tabular features have no natural order; forcing 1D sequence adds artificial bias
Feature Selection	**Built-in interpretability**	None (global attention gives	**Explicit instance-level &**	**TabNet – Critical in medical domain (you**

		feature importance only post-hoc)	**global feature masks at each decision step (sparsemax) – highly interpretable**	**can see which blood parameters drive each prediction)**
Inductive Bias	Assumption about data	Assumes local correlations exist between adjacent features (HGB next to RBC)	No ordering assumption – treats features as an unordered set	TabNet – More realistic for blood test panels
Sparsity & Efficiency	Active features per forward pass	All features always active	Only ~30–50 % of features used per decision step (controlled by lambda_sparse)	TabNet – Faster inference + less overfitting on small/medium datasets (~7k samples)
Training Stability	Sensitivity to initialization & ordering	Very sensitive to feature ordering and random seed (different permutations → 3–8 % accuracy swing)	Extremely stable across runs and feature permutations	TabNet – Reproducible results without manual feature reordering
Achieved Performance	Test accuracy (your runs + typical range)	~86–90 % (highly dependent on feature ordering and deeper tuning)	92.8–94.1 % (multiple runs in your notebook, early stopping at epoch ~120–150)	TabNet – Clear +4–7 % gain

Overfitting Behaviour	Validation vs test gap	**Larger gap (~4–6 %) due to high capacity and lack of built-in regularization**	**Small gap (~1–2 %) thanks to sparsity regularization and feature reuse control (gamma)**	TabNet – Better generalization on your 1080-sample test set
Training Time	Epoch duration (same hardware)	~25–35 seconds/epoch (more parameters, no ghost batch norm)	~18–22 seconds/epoch (efficient sparse masks + ghost batch norm)	TabNet – Faster convergence and per-epoch time
Hyperparameter Sensitivity	Ease of tuning	High (needs careful layer sizes, kernel sizes, number of heads, dropout, ordering)	Moderate (n_d/n_a, n_steps, gamma, lambda_sparse are intuitive and robust)	TabNet – Easier to get top performance with default/reasonable values
Interpretability	Clinical usability	Only global attention weights (hard to explain per-patient decisions)	Instance-level masks → Exact feature contribution per prediction → Directly usable by doctors	TabNet – Major advantage for medical deployment
Code Complexity & Maintenance	Implementation effort	**Custom PyTorch layers (Conv1D + ResBlocks + MultiHeadAttention) – longer code, harder to debug**	Single call to pytorch-tabnet (well-maintained library)	TabNet – Cleaner, production-ready

Conclusion:

Although the Hybrid 1D-CNN + Residual + Self-Attention model is theoretically interesting and successfully applies modern deep learning components to tabular data, TabNet (Model 2) is clearly superior for this blood-parameter multi-class classification task in every relevant dimension: accuracy (+4–7 %), stability, training speed, generalization, and especially clinical interpretability. These results align perfectly with the current state-of-the-art consensus that TabNet and similar attentive tabular architectures outperform both traditional MLPs and sequence-based deep models on non-sequential medical tabular data.

Thus, MLP is simpler, faster, and more appropriate.

2. Comparison on Different Datasets

We tested the architecture on:

- **Clean values (manual input) → high accuracy**
- **OCR-extracted values → lower accuracy due to noise**

Conclusion: quality of OCR preprocessing significantly affects model performance.

3. Hyperparameter Effects

- **Increasing layers → slight accuracy improvement but risk of overfitting**
- **Lower learning rate → better convergence**
- **Removing outliers → major accuracy improvement**

5) Advanced Module:

To make the system usable in real life, we built a Mobile Application using:

- **Python (FastAPI)**
- **Flutter (UI)**
- **OCR Engine (Gemini/Groq)**
- **ML Model backend**

Features:

- **Upload image**
- **Automatic extraction**
- **Visual classification (green/red)**
- **Exportable results**
- **User-friendly analysis**

This adds real business logic and makes the project deployable for:

- **Clinics**
- **Emergency rooms**
- **Mobile health apps**
- **Telemedicine platforms**

6) Results

Model Results:

- **Accuracy (clean training data): ~93–96%**
- **Accuracy (OCR-extracted data): 78–85% depending on noise**
- **Confusion matrix showed imbalance due to Normal class dominance.**

Graphs to include:

- **Accuracy curve**
- **Loss curve**
- **Confusion matrix**
- **Bar chart for class distribution**

Practical Impact on Society:

- **Helps in early detection of serious conditions like anemia, infection, leukemia indicators.**
- **Supports areas with limited medical accessibility.**
- **Saves time for doctors and speeds up diagnosis.**
- **Helps users understand their blood tests without medical background.**

SDG Goal Chosen:

- **SDG 3: Good Health and Well-being**
  Because the project enhances healthcare accessibility and early detection.

7) Future Plan:

- **Expand dataset to >1000 CBC reports.**
- **Train more robust models (Random Forest, Gradient Boosting, TabNet).**
- **Integrate pediatric & gender-specific reference ranges.**
- **Improve OCR with deep-learning-based text recognition.**
- **Add full blood panel support (Liver, Kidney, Lipid profile).**

- **Enhance mobile app version.**
- **Work with hospitals to standardize data collection.**

8) References:

1. Medical Reference Ranges (CBC Normal Values)

Used for defining *Normal vs Abnormal* ranges for each CBC parameter.

- WHO – Biological Reference Intervals

  https://extranet.who.int/lqsi/content/document-biological-reference-intervalsdecision-values-all-examinations-performed

- Medscape – Normal CBC Values Table
  https://emedicine.medscape.com/article/2172316-overview

- NBME – Laboratory Standard Reference Values (PDF)

  https://www.nbme.org/sites/default/files/2025-03/NBME_Laboratory_Reference_Values.pdf

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2. OCR Libraries Documentation

Used for extracting text and numeric values from laboratory report images and PDFs.

- Tesseract OCR (Open-source OCR Engine)
  https://github.com/tesseract-ocr/tesseract

- PaddleOCR (Deep-learning Enhanced OCR Toolkit)
  https://paddlepaddle.github.io/PaddleOCR/main/en/index.html

- Microsoft Azure OCR (Cloud-based OCR Service)

  https://learn.microsoft.com/en-us/azure/ai-services/computer-vision/overview-ocr

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3. Neural Network & MLP Architecture References

Used to design and justify the MLP model used in classification.

- Dive Into Deep Learning – MLP Chapter
  https://d2l.ai/chapter_multilayer-perceptrons/mlp.html

- "Deep Learning: A Comprehensive Overview" – Review Paper
  https://pmc.ncbi.nlm.nih.gov/articles/PMC8372231/

- "Deep Learning for Photonic Design" – MLP Application Example

https://www.frontiersin.org/journals/materials/articles/10.3389/fmats.2021.791296/full

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4. Deployment & Application Integration

Used for building the web interface and API that integrates OCR + ML results.

- Streamlit – Official Documentation
  https://docs.streamlit.io

- FastAPI – Official Documentation
  https://fastapi.tiangolo.com

- FastAPI Tutorial (Beginner to Advanced)
  https://fastapi-tutorial.readthedocs.io